

ITEM 1 BIOLOGY

Scenario

In **Kyoga Sub-county**, many young children who have just stopped breastfeeding are developing swollen bellies, weak bodies, pale skin, and reddish hair. Local elders believe the children are bewitched and are giving them traditional herbs. However, a health check showed that these children eat only cassava porridge and sweet potatoes. Although their parents farm beans, groundnuts, milk, and eggs, they sell all of these foods at the market to earn money. When health workers told the mothers to start feeding the children groundnut paste, some refused, saying that small children's stomachs are too weak to digest such thick foods.

Task

- **Part (a):** Write an essay naming the health condition the children are suffering from and explaining how eating only starchy foods without protein damages a growing child's body.
- **Part (b):** Write an essay explaining why growing children need protein foods, and explain how a child's digestive system actually breaks down proteins to prove the mothers wrong.

ITEM 2: S3 Biology

Topic: Water Movement in Plants and Osmosis

Scenario

A farmer in **Kumi District** put a very large amount of dry artificial fertilizer around the stems of his tomato plants during a sunny week. He then watered the plants using salty water from a deep well. Two days later, even though the soil was wet, all the tomato plants wilted and their leaves drooped. Thinking that insects were attacking the roots, the farmer sprayed a strong chemical on the leaves. By the next morning, all the tomato plants had dried up and died.

Task

- **Part (a):** Write an essay explaining how adding too much fertilizer and salty water caused water to leave the plant roots instead of entering them, leading to wilting.
- **Part (b):** Write an essay explaining why spraying the leaves made the damage worse, and give clear advice on how the farmer should water and fertilize his crops correctly in the future.